

Finite model property for the Guarded Fragment

Lecture 8 (Prague 13.08.2026)

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Plan: the probabilistic method and a warm-up from graph theory, then the guarded construction.

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No object is ever exhibited: a nonempty event certifies a nonempty set.

Reminder: the diagonal Ramsey number $R(n, n)$

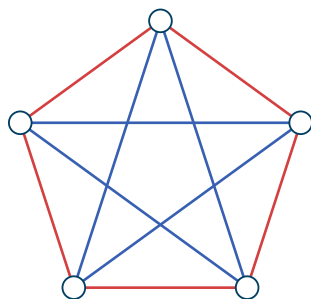
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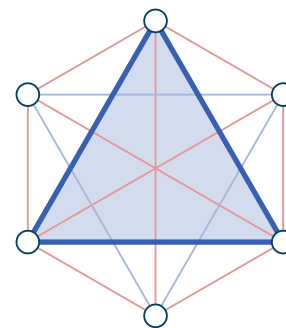
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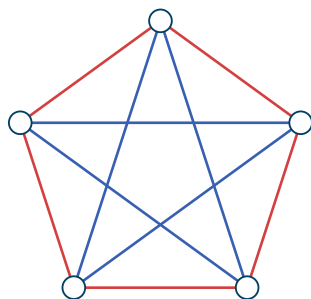


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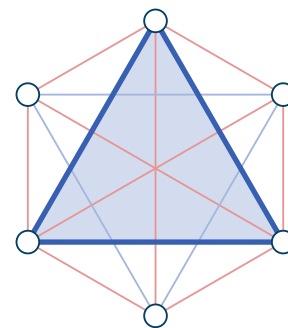
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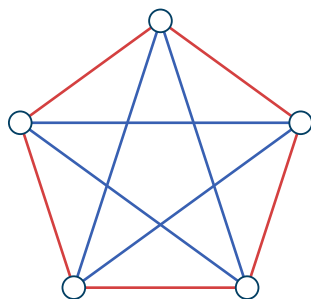
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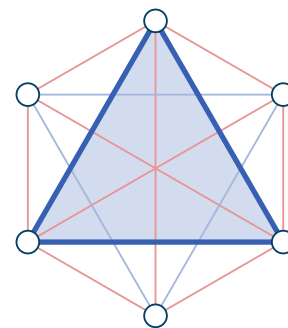
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Upper bounds come from pigeonhole; a **lower** bound needs a colouring of K_N with no monochromatic K_n .

Nobody knows how to write such a colouring down. Erdős: do not write one down.

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So some colouring of K_N has no monochromatic K_n : $R(n, n) > N = \lfloor 2^{n/2} \rfloor$.

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Same four steps, new object. From here on we sample a finite *structure* $\mathfrak{B}$, and *good* no longer means “no monochromatic K_n ” but $\mathfrak{B} \models \varphi$ for a given $\varphi \in \text{GF}$.

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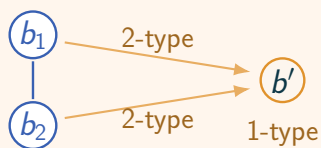
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Where does δ come from?



Let τ be the 3-type of $\bar{a}b_3$ in $\mathfrak{A}$, where $\bar{a}$ matches $b_1 b_2$ and b_3 is its witness.

A fresh b' works as soon as *three* independent draws agree with τ : the 1-type of b' , and the 2-types of (b_1, b') and of (b_2, b') . Hence

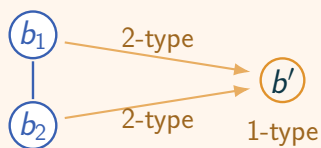
First try: Gurevich–Shelah, when every arity is ≤ 2

Say $\varphi = \forall x_1 x_2 \exists y \psi$ over unary and binary symbols; fix any model $\mathfrak{A} \models \varphi$, and let $\mathfrak{B}$ have n elements.

1. **Sample.** Give each element a 1-type of $\mathfrak{A}$, and each *pair* a random 2-type of $\mathfrak{A}$.
2. **Name the enemy.** $E_{b_1 b_2}$: “the pair $b_1 b_2$ never gets the witness $\mathfrak{A}$ promises it”.
3. **Local estimate.** Every element outside $\{b_1, b_2\}$ is an independent attempt at that witness.
4. **Union bound.** $\Pr[E_{b_1 b_2}] \leq (1 - \delta)^{n-2} \leq e^{-\delta(n-2)}$, and there are only n^2 pairs.

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Let τ be the 3-type of $\bar{a}b_3$ in $\mathfrak{A}$, where $\bar{a}$ matches $b_1 b_2$ and b_3 is its witness.

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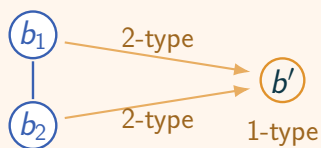
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$\Pr[\mathfrak{B} \not\models \varphi] \leq n^2 e^{-\delta(n-2)} \rightarrow 0$: the Ramsey proof, with “monochromatic” $\rightsquigarrow$ “witnessless”.

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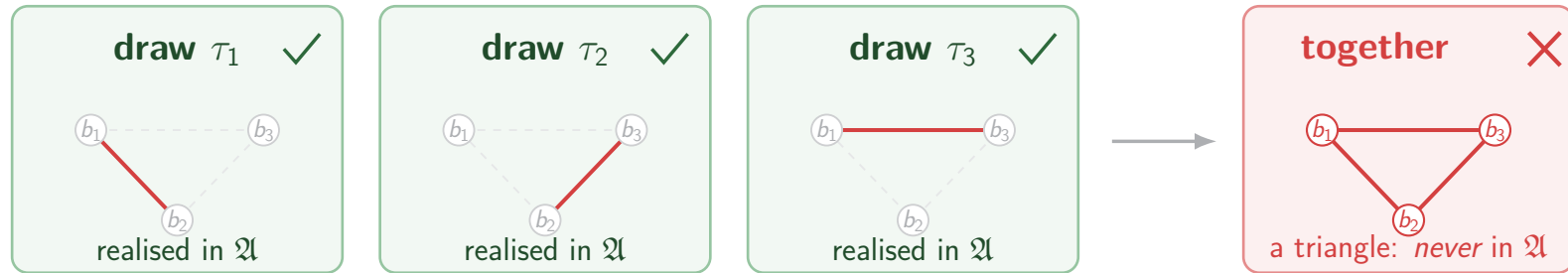
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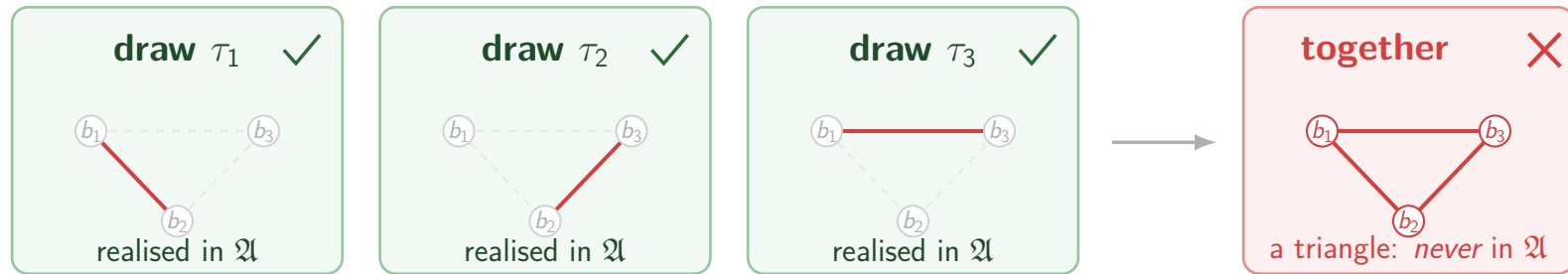
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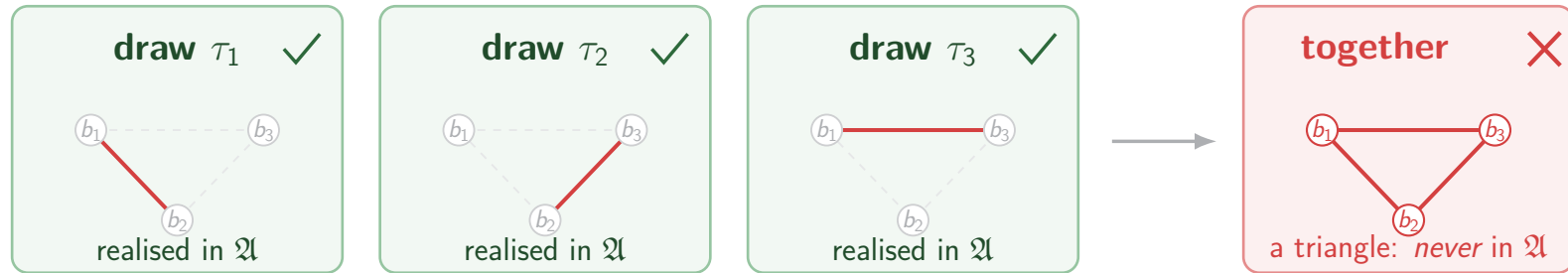


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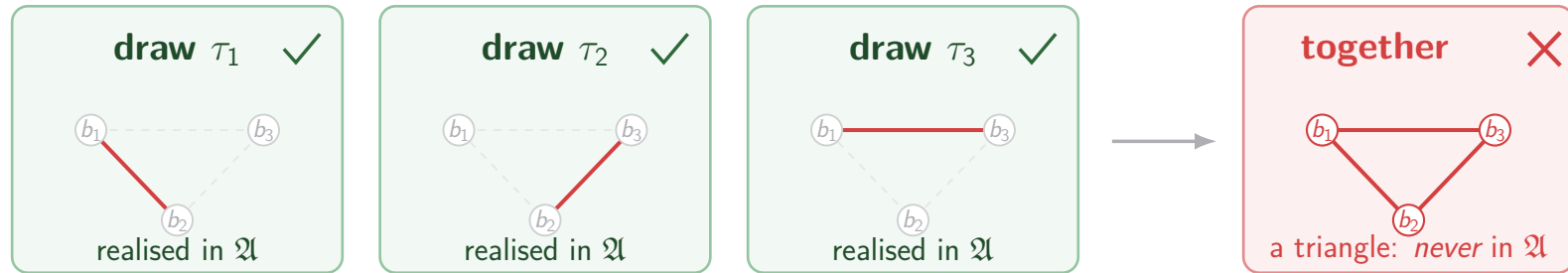
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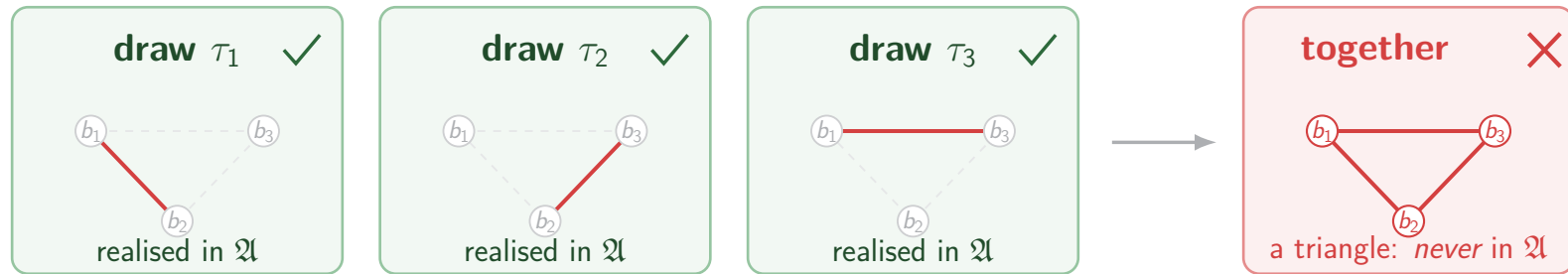
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But that sentence is *not guarded*: no single atom covers x_1, x_2, x_3 .

Guardedness pays the bill

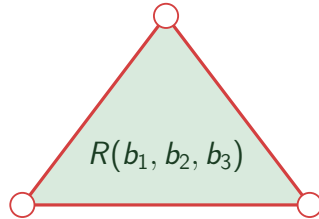
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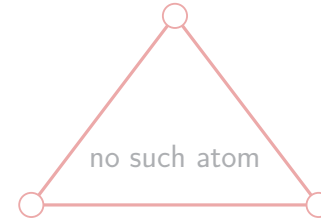
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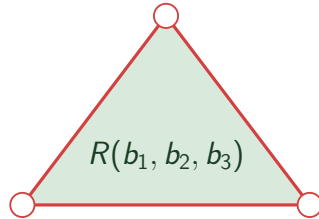


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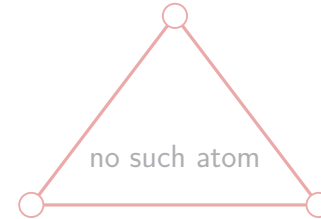
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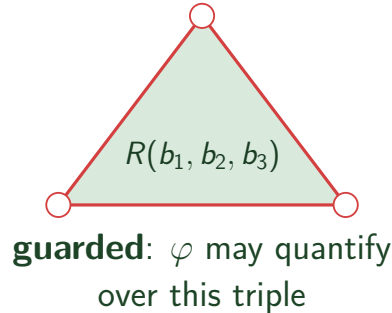
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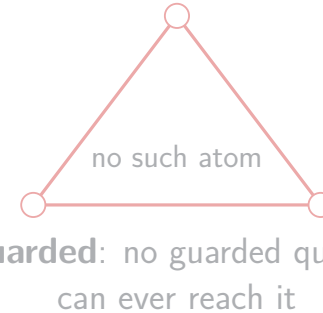
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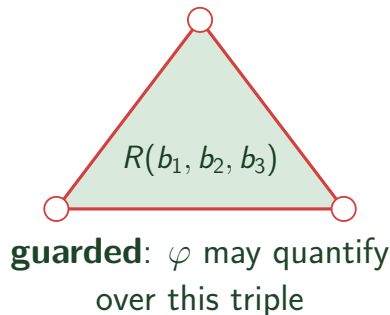
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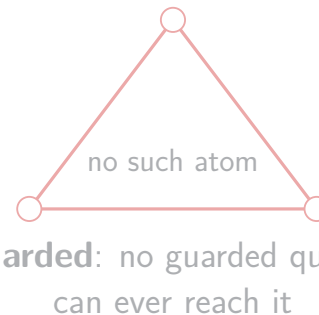
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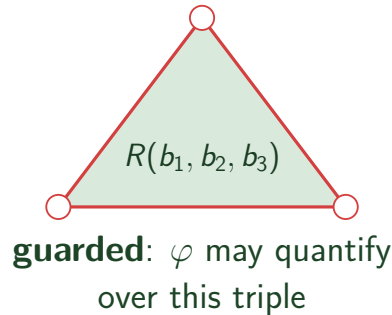
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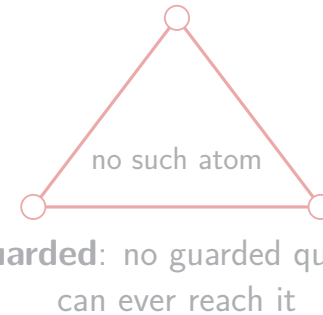
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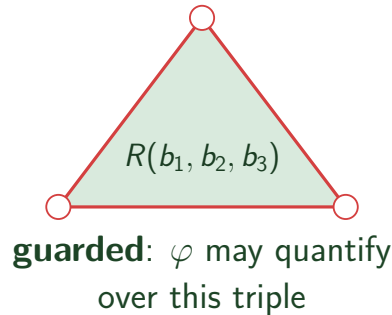
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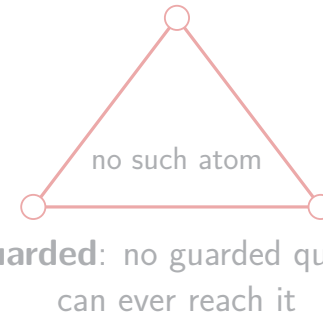
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Conflicts are never prevented – they are made invisible, which is all GF needs.

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Why: a guarded $\bar{b}$ copies some $\bar{a}$ of $\mathfrak{A}$; the witness $\mathfrak{A}$ gives $\bar{a}$ has its type in $\mathcal{T}^*$; now use (ii).

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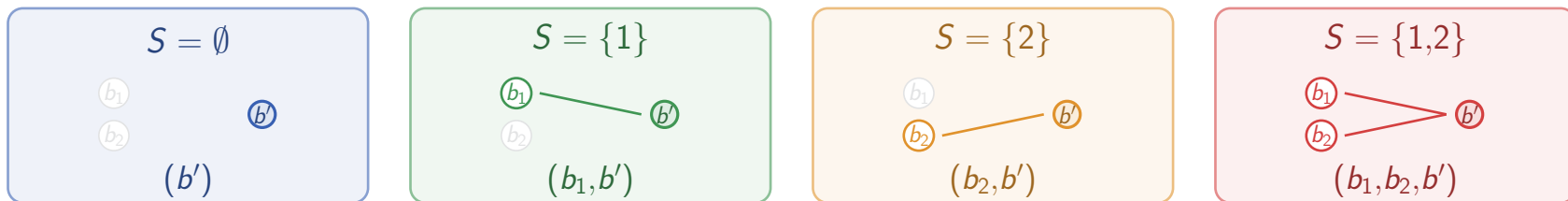
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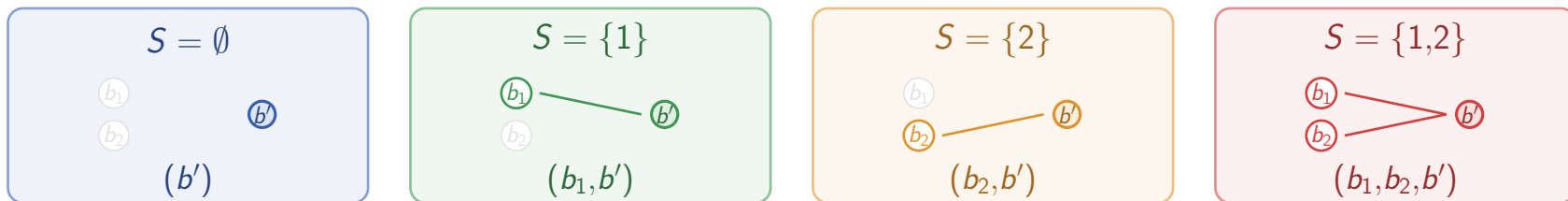
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Each line uses the same three tools as the Ramsey warm-up: independence, $1 - x \leq e^{-x}$, and the union bound.

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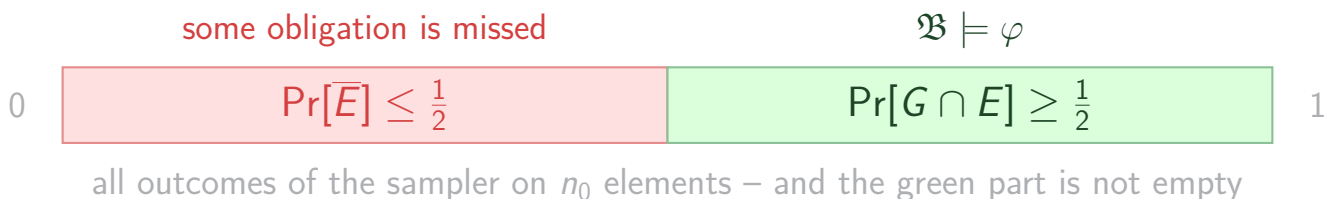
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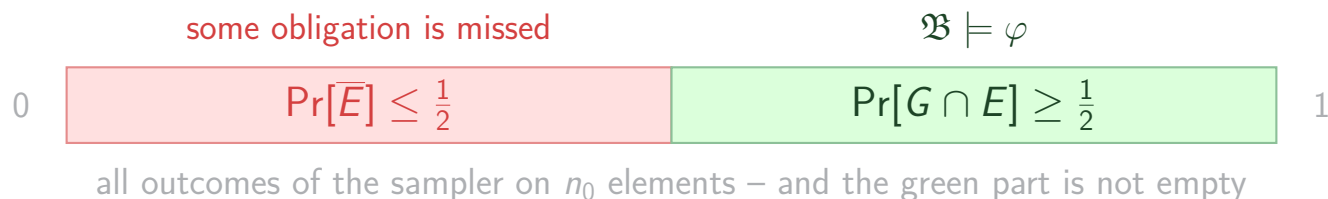
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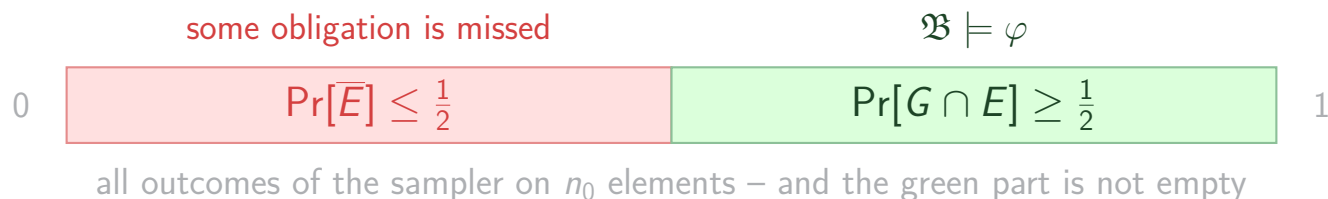


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That outcome is a concrete structure on n_0 elements satisfying φ . GF has the FMP.

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1 Erdős' four-step recipe transfers *verbatim*.

2 Compress $\mathfrak{A}$ into the finite catalogue $\mathcal{T}^*$.

3 Draw one type per tuple; *skip* every clash.

4 Skipping is free: GF cannot see unguarded tuples.

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6 $|\mathcal{T}^*| \leq 2^{2^{O(|\varphi| \log |\varphi|)}}$, and the bound is optimal.

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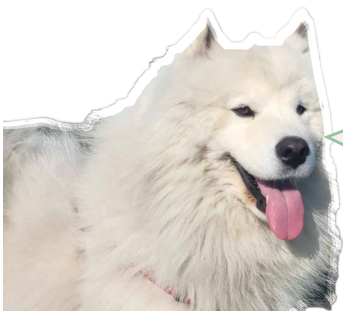
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Thank you!

It's my last day at ESSLLI, so I hope that you enjoyed the course!